



SPACE DESIGN LOAD SUMMARY

2026-09-27

Bedroom 2_3

Level: Level 0 · Internal condition: Double Bedroom - Bedroom 2_3

DESIGN CRITERIA

	Heating	Cooling
Room set point	-50.0 °C	150.0 °C
Outdoor dry bulb	—	—
Outdoor RH (coincident)	—	—
ROOM HUMIDITY		
Humidification set point	—	
lower RH limit		
Dehumidification set point	—	
upper RH limit		

SIZING (TAS DESIGN LOADS)

	Heating	Cooling
Design load	0 W	0 W
Design load per area	0.0 W/m²	0.0 W/m²
Sizing multiplier	not set	not set
Design loads: from Tas sizing — date/currency not recorded		

HEATING

Design day	Leeds_TRY ANN HTG 100% CONDS DB	
	Design day	Full year
Peak sensible load	1,630 W	776 W
Peak hour	14:00–15:00	1 Jan 00:00–01:00 (HOY 1)
Room dry bulb	12.4 °C	12.3 °C
Room resultant	11.3 °C	11.9 °C
Room RH	25 %	75 %
Room humidity ratio	2.2 g/kg	6.7 g/kg
Outdoor dry bulb	—	3.3 °C
Outdoor RH	—	100 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (sensible)	0	105
Equipment (sensible)	0	10
Infiltration / ventilation	-392	-193
Building heat transfer	-461	-373
External conduction — opaque	-567	-224
External conduction — glazing	-209	-101

LATENT COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (latent)	0	77

Zero at both peaks (not listed): Solar, Lighting, Air movement (inter-zone), Air handling unit, Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day	Leeds_TRY ANN CLG 0% CONDS DB=>GRad	
	Design day	Full year
Peak sensible load	2,071 W	1,369 W
Peak hour	00:00–01:00	2 Aug 04:00–05:00 (HOY 5117)
Room dry bulb	19.3 °C	17.2 °C
Room resultant	20.8 °C	18.3 °C
Room RH	100 %	100 %
Room humidity ratio	14.1 g/kg	12.3 g/kg
Outdoor dry bulb	—	15.2 °C
Outdoor RH	—	94 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Solar	0	11
Occupancy (sensible)	105	105
Equipment (sensible)	10	10
Infiltration / ventilation	-2	-43
Building heat transfer	1,394	1,050
External conduction — opaque	602	296
External conduction — glazing	-38	-60

LATENT COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (latent)	77	77

Zero at both peaks (not listed): Lighting, Air movement (inter-zone), Air handling unit, Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	SAM.Analytical.Tas	SAM.Analytical.Tas
Read into model	27 Sep 2026 20:42	27 Sep 2026 20:42
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year, and its hour of the year (HOY 1 = 1 Jan 00:00–01:00, HOY 8760 = 31 Dec 23:00–24:00).
Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.